

Jeffrey Chong Loh Loy Fatt

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Experience

Tan Tock Seng Hospital | Contract

Full Stack Engineer

December 2025 - Present

- Led the overhaul of AutoVA, a digital health app, refactoring a PoC into a production-ready system utilizing a FastAPI backend, React frontend, Redis and deployed on AWS (EC2, RDS, S3).
- Engineered a high-precision pose monitoring system utilizing MediaPipe and OpenCV with ArUco markers, optimized inference for real-time clinical accuracy.
- Converted clinical and operational needs into clear user stories and delivery milestones, enabling parallel team development and reducing ambiguity.
- Built an internal OCR-based extraction tool (PaddleOCR + Streamlit) to transform raw operational documents into structured data, facilitating downstream advanced data analytics and trend monitoring increasing efficiency by 50%.

Rovisys Asia Company Pte Ltd | Full-time

Lead Software Engineer

December 2023 – May 2025

- Led the design and implementation of Semiconductor Manufacturing Execution System (MES) solutions in C#, leveraging Critical Manufacturing MES to optimize workflows, integrate systems, and meet client-specific requirements.
- Authored comprehensive, industry-standard documentation for MES architectures, workflows, and codebases, ensuring maintainability and knowledge transfer across teams
- Managed deployment of solutions to client environment, ensuring seamless integration and minimal disruption to operations.
- Collaborated cross-functionally in Agile Scrum environment with stakeholders to identify requirements, define project roadmaps, and deliver solutions that drive measurable business value.
- Mentored a team of 3 junior developers, fostering skill development and promoting best practices in coding, system design, and documentation.

Software Engineer

December 2021 – November 2023

- Enhanced and maintained a large-scale analytics platform built with Angular, Django, and C#, sustaining high availability and performance under continuous change.
- Built a custom C# SQL Server Express backup tool, guaranteeing 100% data integrity for operational data stores.
- Designed SQL Server schemas and developed T-SQL stored procedures to support PowerBI dashboards and automated daily ETL ingestion workflows.
- Created an internal automation tool to accelerate dashboard provisioning for new deployment sites.

Education

National University of Singapore - MTech in Artificial Intelligence Systems (Part-time) January 2025 – December 2026

University of Wollongong - BSc Computer Science (Big Data) (Part-time) April 2022 – September 2024

Personal Projects

Thrive: Personalized Health Risk Assessment & AI Chatbot

React · FastAPI · Docker · TailwindCSS

GitHub: <https://github.com/JC-prog/Thrive-Personalized-Health-Chatbot>

- Built a full-stack healthcare application to assess heart disease and diabetes risk using lifestyle and health data.
- Integrated ML-based risk prediction and an LLM-powered chatbot (OpenAI API) with embedding-based contextual retrieval.
- Exposed inference and conversational workflows via FastAPI APIs, containerized with Docker for deployment readiness.

F1 Regulation RAG

Github: <https://github.com/JC-prog/F1-Regs-Rag>

- Developed a high-fidelity retrieval system to query 100+ pages of dense FIA technical standards using LangChain, LangGraph, and Gemini API.
- Optimized the system by transitioning from Naive RAG to a Metadata-Aware architecture by metadata filtering across Technical, Sporting, and financial categories to eliminate contextual drift.

Skill Summary

Programming: Python (FastAPI, Django), C# (.NET 8, EF Core), Java, SQL (T-SQL), Bash/Shell Scripting.

AI Engineering: LLM Integration, RAG, Computer Vision (MediaPipe, OpenCV, ArUco), ASR (Whisper), OCR (PaddleOCR),

AI/ML Libraries: PyTorch, TensorFlow, Scikit-learn, YOLOv12, Computer Vision (OpenCV, MediaPipe), RAG (LangGraph, LangChain, Vector DBs), LLM Integration (Gemini/OpenAI API).

AI-Assisted Development: Claude Code, Gemini, Antigravity, ChatGpt

Infra & DevOps: Docker, CI/CD (GitLab, Azure DevOps, SonarQube), AWS (EC2, S3, RDS), Prometheus & Grafana.